

# WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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## 1. INTRODUCTION

**Theorem 1.1.** *Let  $R$  be a normalised root system in  $\mathbb{R}^N$ ,  $N \geq 2$ , and let  $G$  be the corresponding finite reflection group. Let  $k : R \rightarrow \mathbb{R}_+$  be  $G$ -invariant multiplicity function. Let  $R_{\text{Dunkl},j}$  be the  $j$ -th Dunkl-Riesz transform corresponding to  $R$  and  $k$ . Let  $w$  be the Dunkl weight corresponding to  $R$  and  $k$ . Let  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ .*

- (i) *If  $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$ ,  $p > N$ , then  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ . Furthermore,*

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

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(ii) If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$ ,  $p > N$ , then  $f \in \dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{p,R,k}^{-1} \|f\|_{\dot{W}^{\frac{N}{p},p}(\mathbb{R}^N)};$$

(iii) If  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ , then  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

(iv) If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ , then  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_{R,k}^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

## 2. PRELIMINARIES

**2.1. Harmonic analysis in the Dunkl setting.** Let  $\mathbb{R}^N$  be the Euclidean space equipped with the usual scalar product  $\langle \cdot, \cdot \rangle$  and the induced norm  $\|\cdot\|$ . For a nonzero vector  $v \in \mathbb{R}^N$ , the reflection  $\sigma_v$  with respect to the hyperplane  $v^\perp$  orthogonal to  $v$  is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set  $R \subset \mathbb{R}^N \setminus \{0\}$  is called a root system if  $\sigma_v(R) = R$  for every  $v \in R$ . A root system  $R$  is called a normalized reduced root system if  $\|v\|^2 = 2$  for every  $v \in R$ . The finite group  $G$  generated by the reflections  $\{\sigma_v\}_{v \in R}$  is called the Weyl group (reflection group) of the root system  $R$ . Let  $k : R \rightarrow [0, \infty)$  be a multiplicity function, which is a  $G$ -invariant function fixed throughout the paper.

Define  $d(x, y) = \min_{g \in G} |x - gy|$ . Recall from [7, Chapter VII, proof of Theorem 2.12] that  $d(x, y) = |x - y|$  iff  $x$  and  $y$  belong to the same chamber.

Define the Dunkl measure  $dw$  on  $\mathbb{R}^N$  by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where  $dx$  stands for the Lebesgue measure on  $\mathbb{R}^N$ . Let  $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$  be the Euclidean ball centred at  $x$  with radius  $r > 0$ . Let  $\mathbf{N} := N + \sum_{v \in R} k(v)$ . Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on  $N, R$  and  $k$ . It follows from (1) that there exists a constant  $C \geq 1$  such that for every  $x \in \mathbb{R}^N$ ,

$$(2) \quad C^{-1} \left( \frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left( \frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers  $\mathbf{N}$  and  $N$  are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators  $T_\xi$  are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where  $\partial_\xi f$  denotes the directional derivative of  $f$  in the direction  $\xi$ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where  $\Delta_{\mathbb{R}^N}$  is the Laplacian on  $\mathbb{R}^N$  and  $\delta_v$  is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by  $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$ ,  $j = 1, 2, \dots, N$ .

## 2.2. Weyl chamber as simplicial cone.

**Definition 2.1.** *A connected component of the set*

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

*is called an open Weyl chamber. Its closure is called a Weyl chamber.*

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

**Theorem 2.2.** *Let  $\Gamma$  be a chamber. There exists a linear isomorphism  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  such that  $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$  (with  $N_1 + N_2 = N$ ).*

## 2.3. System of dyadic cubes.

**Notation 2.3.** *Let*

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

*be the standard system of dyadic cubes over  $\mathbb{R}^N$ . Here each dyadic cube in  $\mathbb{D}_k$  is of the shape*

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

*For every dyadic cube  $Q \in \mathbb{D}$ , we denote by  $c_Q$  and  $\ell(Q)$  its center and side-length, respectively.*

**2.4. Nearly weakly orthonormal sequences.** Rochberg and Semmes [12] introduced the notion of *nearly weakly orthonormal (NWO)* sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [5, 9, 11, 3, 4, 10, 12, 14, 13]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [9, Section 13] for a complete proof).

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

**Definition 2.4.** Let  $w \in A_2(\mathbb{R}^N)$ . Let  $c > 0$  and  $2 < r \leq \infty$ . A sequence of functions  $\{e_Q\}_{Q \in \mathbb{D}}$  is called  $(r, c)$ -NWO if

- (1)  $\text{supp}(e_Q) \subseteq cQ$  for every  $Q \in \mathbb{D}$ ;
- (2)  $\|e_Q\|_{L_r(\mathbb{R}^N, w(x)dx)} \leq w(Q)^{\frac{1}{r} - \frac{1}{2}}$  for every  $Q \in \mathbb{D}$ ;

**Lemma 2.5.** Let  $w \in A_2(\mathbb{R}^N)$ . Let  $0 < p < \infty$ . Suppose that  $A$  is an integral operator on  $L_2(\mathbb{R}^N, w(x)dx)$  with integral kernel

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where  $\{e_Q\}_{Q \in \mathbb{D}}$  and  $\{f_Q\}_{Q \in \mathbb{D}}$  are  $(r_1, c_1)$ -NWO and  $(r_2, c_2)$ -NWO sequences, respectively. We have

$$\begin{aligned} \|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty}, \\ \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, w(x)dx))} \left( A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)) \right) &\leq \\ &\leq C_{w,r_1,c_1,r_2,c_2,p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left( \{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

*Proof.* The assertion follows from Corollary 8.2 and Lemma 8.7 in [9].  $\square$

### 3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

**Theorem 3.1.** Let  $\Gamma$  be a chamber. For all  $x \in \Gamma$  and for all  $r > 0$ , we have

$$w(B(x, r) \cap \Gamma) \geq c_\kappa w(B(x, r)).$$

In particular,  $(\Gamma, |\cdot|, w(x)dx)$  is a doubling metric measure space.

**Notation 3.2.** Let  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ . Set

$$w_\theta(x) = \prod_{A \subset \{1, \dots, N_1\}} \left( \sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

**Lemma 3.3.** For every  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  and for every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ , we have

$$\int_{B(x,r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq c_{N,\theta}^{(1)} r^{N_1} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

*Proof.* Obviously,

$$B(x, r) \subset \left( \times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} &\int_{B(x,r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq \\ &\leq \int_{\left( \times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_\theta(y_1, \dots, y_{N_1}) dy = \\ &= \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_\theta(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ &\leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ &= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \end{aligned}$$

$$\leq w_\theta((x_1, \dots, x_{N_1}) + r\mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2})r^{N_2}.$$

□

**Lemma 3.4.** *For every  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  and for every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ , we have*

$$\int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq c_{N,\theta}^{(2)} r^N w_\theta((x_1, \dots, x_{N_1}) + r\mathbf{1}_{N_1}).$$

*Proof.* Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\ & = w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^{N_2} \geq \\ & \geq w_\theta((x_1, \dots, x_{N_1}) + r\mathbf{1}_{N_1}) \times (\frac{r}{2N})^{N_2} \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}. \end{aligned}$$

□

**Lemma 3.5.** *There exists  $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$  such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

*Proof.* For every  $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$  and for every  $v \in R$ , we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone  $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let  $A_v$  be the set of those  $1 \leq k \leq N_1$  with  $((\Phi^{-1})^*(v))_k$  being strictly positive. We, obviously, have

$$c_{\Phi,v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi,v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi,v}^{-\kappa(v)} \left( \sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi,v}^{\kappa(v)} \left( \sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. □

*Proof of Theorem 3.1.* The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. □

## 4. FIRST ROCHBERG-SEMMES BASED ESTIMATE

**Theorem 4.1.** *Let  $N \geq 2$ ,  $1 \leq N_1 \leq N$  and  $N_2 = N - N_1$ . Let  $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . Let  $F$  be a face of  $\Omega$  (not necessarily of maximal dimension, possibly of zero dimension). Let  $\text{id} \neq \sigma : \mathbb{R}^N \times \mathbb{R}^N$  be the reflection with respect to  $F$ . Let  $\theta : 2^N \rightarrow \mathbb{R}_+$ . Suppose that  $K : \Omega \times \Omega \rightarrow \mathbb{C}$  satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{|x - \sigma y| \cdot w_\theta(x, |x - y|)}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

*Let  $V$  be an integral operator on  $L_2(\Omega, w_\theta(x)dx)$  with integral kernel  $K$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$  vanishing on  $F$ , we have*

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, w_\theta(x)dx))} \leq c_{K, \theta} \|\nabla f\|_{L_N(\Omega)}.$$

**Theorem 4.2.** *Suppose we are the setting of Theorem 4.1. If  $0 \leq \phi_m \leq 1$  for  $m \geq 1$  are in  $C_c^\infty(\mathbb{R}^N)$  and  $\text{supp}(1 - \phi_m)$  increases to  $\mathbb{R}^N$ , then*

$$\text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Omega, w_\theta(x)dx))} \left( M_{\phi_m} \cdot M_f V \cdot M_{\phi_m} - M_f V, (\mathcal{L}_{N, \infty})_0(L_2(\Omega, w_\theta(x)dx)) \right) \rightarrow 0$$

*as  $m \rightarrow \infty$ .*

## 4.1. Proof of Theorem 4.1.

**Lemma 4.3.** *There exists a smooth function  $\psi$  supported on  $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$  such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

*Proof.* Choose  $\rho \in C_c^\infty(\mathbb{R}^N)$  such that  $\rho = 1$  on  $(-\frac{1}{2}, \frac{1}{2})^N$  and  $\rho = 0$  outside  $(-1, 1)^N$ . Set  $\psi(z) = \rho(z) - \rho(2z)$ . Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every  $z \neq 0$ , we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every  $x$ , there exists a unique  $Q \in \mathbb{D}_j$  containing  $x$ . For this  $Q$ , we have  $\ell(Q) = 2^{-j}$ . We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left( \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If  $x \in Q$  and  $\psi(\frac{x - y}{\ell(Q)}) \neq 0$ , then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have  $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$  and  $|x - y|_\infty \leq \ell(Q)$ . Thus,  $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$ . In other words,  $y \in 3Q$ . Thus,

$$\chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

This completes the proof.  $\square$

**Lemma 4.4.** *If  $Q \in \mathbb{D}$  is such that  $Q \subset \Omega$ ,  $x \in Q$  and  $y \in 3Q \cap \Omega$ , then*

$$\frac{|x - y|}{|x - \sigma y|} \leq \frac{3N\ell(Q)}{\text{dist}(c_Q, F)}.$$

*Proof.* Suppose first that  $3Q \subset \Omega$ . We have

$$\begin{aligned} |x - \sigma y| &\geq |x - \sigma y|_\infty \geq |c_Q - \sigma c_Q|_\infty - |x - c_Q|_\infty - |y - c_Q|_\infty = \\ &= 2\text{dist}_{l_\infty^N}(c_Q, F) - |x - c_Q|_\infty - |y - c_Q|_\infty. \end{aligned}$$

Since  $3Q \subset \Omega$ , it follows that  $\text{dist}_{l_\infty^N}(c_Q, F) \geq \frac{3}{2}\ell(Q)$ . Since  $x \in Q$  and  $y \in 3Q$ , it follows that  $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$  and  $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$ . Thus,

$$\begin{aligned} |x - \sigma y| &\geq 2\text{dist}_{l_\infty^N}(c_Q, F) - 2\ell(Q) = \\ &= \frac{2}{3}\text{dist}_{l_\infty^N}(c_Q, F) + \frac{4}{3}(\text{dist}_{l_\infty^N}(c_Q, F) - \frac{3}{2}\ell(Q)) \geq \\ &\geq \frac{2}{3}\text{dist}_{l_\infty^N}(c_Q, F) \geq \frac{2}{3}N^{-\frac{1}{2}}\text{dist}(c_Q, F). \end{aligned}$$

On the other hand,

$$|x - y| \leq N^{\frac{1}{2}}|x - y|_\infty \leq N^{\frac{1}{2}}(|x - c_Q|_\infty + |y - c_Q|_\infty) \leq 2N^{\frac{1}{2}}\ell(Q).$$

Thus,

$$\frac{|x - y|}{|x - \sigma y|} \leq \frac{3N\ell(Q)}{\text{dist}(x, F)}.$$

If cube  $Q$  touches the boundary of  $\Omega$ , then

$$\frac{|x - y|}{|x - \sigma y|} \leq 1 = \frac{2\ell(Q)}{\text{dist}(c_Q, F)}.$$

□

**Lemma 4.5.** *In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

- (i)  $e_{k, l, Q}$  is supported in  $Q$  and  $f_{k, l, Q}$  is supported in  $3Q$  for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ ;
- (ii) for every  $k, l \in \mathbb{Z}^N$  and for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have

$$\|e_{k, l, Q}\|_{L_{2N}(\mathbb{R}^N, w_\theta(x)dx)} = \|f\|_{L_{2N}(Q, w_\theta(x)dx)};$$

- (iii) for every  $k, l \in \mathbb{Z}^N$  and for every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have

$$\|f_{k, l, Q}\|_{L_\infty(\mathbb{R}^N)} \leq c_{K, \kappa} \frac{\ell(Q)}{\text{dist}(c_Q, F)};$$

*Proof.* For every  $x, y \in \Omega$ , we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x - y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x - y}{\ell(Q)}\right) = \end{aligned}$$

For every  $Q \in \mathbb{D}$  with  $3Q \subset \Omega$ , define the map

$$K_Q : [-\frac{1}{2}, \frac{1}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a - b), \quad a \in [-\frac{1}{2}, \frac{1}{2}]^N, \quad b \in [-\frac{3}{2}, \frac{3}{2}]^N.$$

If  $Q \in \mathbb{D}$  is such that  $3Q \not\subset \Omega$ , then the definition of  $K_Q$  is the same,  $a$  also runs through  $[-\frac{1}{2}, \frac{1}{2}]^N$ , but the parallelepiped  $b$  runs through is slightly smaller. **to co-authors: if you wish, specify the domain of  $b$  also.** In what follows, we lighten the notations by pretending that  $b$  is always defined on  $[-\frac{3}{2}, \frac{3}{2}]^N$ .

It follows from the assumption on  $K$  and Lemma 4.4 that

$$\|K_Q\|_{C^{2N+2}([-\frac{3}{2}, \frac{3}{2}]^N \times [-\frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)}.$$

Hence, there exists an extension  $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$  such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left( \frac{c_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\text{supp}(e_{k,l,Q}) \subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q,$$

$$\|e_{k,l,Q}\|_{L_{2N}(\mathbb{R}^N, w_\theta(x)dx)} = \|f\|_{L_{2N}(Q, w(x)dx)},$$

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} = (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta}}{w_\theta(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)}.$$

□

**Lemma 4.6.** *For every  $Q \in \mathbb{D}$  with  $Q \subset \Omega$ , we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

*Proof.* We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if  $x \in Q$ , then  $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$  and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof.  $\square$

**Lemma 4.7.** *In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x) dx))} \leq c_{K,\theta} \left\| \left\{ m(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}} \right\|_{N,\infty}.$$

*Proof.* Let  $A_{k,l}$  be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.5, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x) dx))} \leq \\ & \leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x) dx))}. \end{aligned}$$

By Lemma 4.5, the sequences

$$\left\{ w_\theta(Q)^{\frac{1}{2N}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_{2N}(Q, w_\theta(x) dx)}} \right\}_{Q \in \mathbb{D}}, \quad \left\{ c_{K,\theta}^{-1} w_\theta(Q)^{\frac{1}{2}} \cdot \frac{\text{dist}(c_Q, F)}{\ell_Q} f_{k,l,Q} \right\}_{Q \in \mathbb{D}}$$

are  $(2N, 1)$ -NWO and, respectively,  $(\infty, 3)$ -NWO in the sense of Definition 2.4.

Using Lemma 2.5, we conclude that

$$\begin{aligned} & \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x) dx))} \leq \\ & \leq c_{\theta, 2N, 1, \infty, 3, N} c_{K,\theta} \left\| \left\{ w_\theta(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q, w_\theta(x) dx)} \cdot \frac{\ell_Q}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} & \|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, w_\theta(x) dx))} \leq \\ & \leq \frac{N}{N-1} c_{\theta, 2N, 1, \infty, 3, N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \end{aligned}$$

$$\times \left\| \left\{ w_\theta(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q, w_\theta(x)dx)} \cdot \frac{\ell_Q}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}, Q \subset \Omega} \right\|_{N, \infty}.$$

By Lemma 4.6,

$$w_\theta(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q, w_\theta(x)dx)} \leq c_\theta^{\frac{1}{2N}} m(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof.  $\square$

**Lemma 4.8.** *Let  $Q$  and  $F$  be as in Theorem 4.1. For every  $f \in C_c^\infty(\mathbb{R}^N)$  vanishing on  $F$ , we have*

$$\left\| \left\{ m(I)^{-1} \|f\|_{L_1(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}, Q \subset \Omega} \right\|_{N, \infty} \leq c_N \|\nabla f\|_{L_N(\Omega)}.$$

*Proof.* Let  $\mathcal{M}$  be the Hardy–Littlewood maximal operator on  $\mathbb{R}^N$ . By [?, Theorem 3], we have **NO! we must handle this difficulty!**

$$|f(x)| \leq c_N \text{dist}(x, F) \cdot (\mathcal{M}(|\nabla f|))(x), \quad x \in \Omega.$$

We have

$$\begin{aligned} \text{dist}(x, F) &\leq |x - c_Q| + \text{dist}(c_Q, F) \leq \frac{d^{\frac{1}{2}}}{2} \ell(Q) + \text{dist}(c_Q, F) \leq \\ &\leq (N^{\frac{1}{2}} + 1) \text{dist}(c_Q, F) \leq (N^{\frac{1}{2}} + 1) |c_Q - \sigma c_Q|, \quad x \in Q. \end{aligned}$$

Thus,

$$|f(x) - f(gx)| \leq c_N \cdot (N^{\frac{1}{2}} + 1) \cdot |c_Q - \sigma c_Q| \cdot (\mathcal{M}(|\nabla f|))(x).$$

This immediately implies

$$\frac{1}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c_\Phi'' \cdot \left( \|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right)$$

for every  $I \in \mathbb{D}_\Gamma$ . Hence,

$$\begin{aligned} m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} &\leq \\ &\leq c_\Phi''' \cdot m(I)^{\frac{1}{N}-1} \left( \|\mathcal{M}(|\nabla f|)\|_{L_1(I)} + \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right) \end{aligned}$$

for every  $I \in \mathbb{D}_\Gamma$ . Thus,

$$\begin{aligned} &\left\| \left\{ m(I)^{-1} \|f - f \circ g\|_{L_1(I)} \cdot \frac{d_I}{|c_I - gc_I|} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \leq \\ &\leq c_\Phi''' \frac{N}{N-1} \left( \left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} + \right. \\ &\quad \left. + \left\| \left\{ m(I)^{\frac{1}{N}-1} \|\mathcal{M}(|\nabla f|)\|_{L_1(gI)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{N, \infty} \right) \leq \\ &\stackrel{L.A.1}{\leq} c_\Phi''' \frac{N}{N-1} c_N \left( \|\mathcal{M}(|\nabla f|)\|_{L_N(\Gamma)} + \|\mathcal{M}(|\nabla f|)\|_{L_N(g\Gamma)} \right) \leq \\ &\leq c_\Phi''' \frac{N}{N-1} c_N \|\mathcal{M}\|_{L_N(\mathbb{R}^N) \odot} \left( \|\nabla f\|_{L_N(\Gamma)} + \|\nabla f\|_{L_N(g\Gamma)} \right). \end{aligned}$$

$\square$

**Lemma 4.9.** *Let  $Q$  and  $F$  be as in Theorem 4.1. For every  $f \in C_c^\infty(\mathbb{R}^N)$  vanishing on  $F$ , we have*

$$\left\| \left\{ m(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}, Q \subset \Omega} \right\|_{N, \infty} \leq c_N \|\nabla f\|_{L_N(\mathbb{R}^N)}.$$

*Proof.* We obviously have

$$\|f\|_{L_{2N}(Q)} \leq \left\| f - \frac{1}{m(Q)} \int_Q f \right\|_{L_{2N}(Q)} + m(Q)^{\frac{1}{2N}-1} \|f\|_{L_1(Q)}.$$

By Sobolev inequality,

$$\|f\|_{L_{2N}(Q)} \leq c_N \|\nabla f\|_{L_{\frac{2N}{3}}(Q)} + m(Q)^{\frac{1}{2N}-1} \|f\|_{L_1(Q)}.$$

By the triangle inequality,

$$\begin{aligned} & \left\| \left\{ m(Q)^{-\frac{1}{2N}} \|f\|_{L_{2N}(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}_{\Omega}} \right\|_{N, \infty} \leq \\ & \leq \frac{2N}{N-1} c_N \left\| \left\{ m(Q)^{-\frac{1}{2N}} \|\nabla f\|_{L_{\frac{2N}{3}}(Q)} \right\}_{Q \in \mathbb{D}_{\Omega}} \right\|_{N, \infty} + \\ & + \frac{N}{N-1} \left\| \left\{ m(Q)^{-1} \|f\|_{L_1(Q)} \cdot \frac{\ell(Q)}{\text{dist}(c_Q, F)} \right\}_{Q \in \mathbb{D}_{\Omega}} \right\|_{N, \infty}. \end{aligned}$$

The assertion follows now from Lemma A.1 (for the first summand) and Lemma 4.8 (for the second summand).  $\square$

*Proof of Theorem 4.1.* The assertion follows by combining Lemmas 4.7 and 4.9.  $\square$

#### 4.2. Proof of Theorem 4.2.

**Lemma 4.10.** *Let  $f, \phi \in C_c^\infty(\mathbb{R}^N)$ . If  $0 \leq \phi \leq 1$ , then*

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_f V \phi, (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K, \kappa, \phi, r} \times \\ & \times \text{dist}_{l_{N, \infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - g c_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N, \infty})_0(\mathbb{D}_\Gamma) \right), \\ & \text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_f V M_\phi, (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq c_{K, \kappa, \phi, r} \times \\ & \times \text{dist}_{l_{N, \infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - g c_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N, \infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

*Proof.* We only prove the second inequality. The proof of the first one is similar, but simpler.

Let  $A_{k, l}$  be as in the proof of Lemma 4.7. We have

$$M_f V - M_f V M_\phi = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k, l} M_{1-\phi}.$$

By the triangle inequality,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left( M_f V - M_f V M_\phi, (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq \frac{N}{N-1} \sum_{k, l \in \mathbb{Z}^N} \left( (1 + |k|^2 + |l|^2)^{-N-1} \times \right. \\ & \left. \times \text{dist}_{\mathcal{L}_{N, \infty}(L_2(\Gamma, w(x)dx))} \left( A_{k, l} M_{1-\phi}, (\mathcal{L}_{N, \infty})_0(L_2(\Gamma, w(x)dx)) \right) \right). \end{aligned}$$

By Lemma 4.5, the sequences

$$\left\{ w(I)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,I}}{\|f\|_{L_r(I,w(x)dx)}} \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ c_{K,\kappa,\Phi}^{-1} w(I)^{\frac{1}{2}} \cdot \frac{|c_I - gc_I|}{d_I} f_{k,l,I} \cdot (1-\phi) \right\}_{I \in \mathbb{D}_\Gamma}$$

are  $(r, 1)$ -NWO and, respectively,  $(\infty, 3)$ -NWO in the sense of Definition ?? . Obviously,  $e_{k,l,I} \neq 0$  only if  $I \cap \text{supp}(f) \neq \emptyset$  and  $f_{k,l,I} \cdot (1-\phi) \neq 0$  only if  $I \cap \text{supp}(f) \neq \emptyset$  and  $3I \cap \text{supp}(1-\phi) \neq \emptyset$ . Using Lemma ?? , we conclude that

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( A_{k,l}(1-\phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi,\kappa,r,1,\infty,3,N} c_{K,\kappa,\Phi} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I w(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I,w(x)dx)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

By Lemma 4.6,

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I,w(x)dx)} \leq c_{\kappa,\Phi}^{\frac{1}{r}} m(I)^{-\frac{1}{r}} \|f\|_{L_r(I)}, \quad I \in \mathbb{D}_\Gamma.$$

Thus,

$$\begin{aligned} & \text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( A_{k,l}(1-\phi), (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \leq \\ & \leq c_{\Phi,\kappa,r,1,\infty,3,N} c_{K,\kappa,\Phi} c_{\kappa,\Phi}^{\frac{1}{r}} \times \\ & \times \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

Combining this estimate with the one in the first paragraph **assign label and refer to it** yields the second inequality.  $\square$

**Lemma 4.11.** *For a bounded Borel  $A \subset \mathbb{R}^N$ , we have*

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq c_N m(A)^{\frac{1}{N}}.$$

*Proof.* For a given  $\epsilon > 0$ , choose  $j$  (depending on  $\epsilon$ ) and finite set  $B \subset \mathbb{D}_j$  such that  $A \subset \cup_{Q' \in B} Q'$  and such that  $m((\cup_{Q' \in B} Q') \setminus A) < \epsilon$ . For  $n \geq m$ , we have (here, we use the notation  $Q_0 = [0, 1)^N$ )

$$\begin{aligned} & \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset Q' \text{ for some } Q' \in B}} \right\|_{l_{N,\infty}(\mathbb{D})} = \\ & = \left( \frac{|B|}{2^j} \right)^{\frac{1}{N}} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset Q_0}} \right\|_{l_{N,\infty}(\mathbb{D})} = c_N \cdot m^{\frac{1}{N}}(\cup_{Q' \in B} Q') \leq c_N \cdot (m(A) + \epsilon)^{\frac{1}{N}}. \end{aligned}$$

Passing  $n \rightarrow \infty$  and, after that, passing  $\epsilon \rightarrow 0$ , we complete the proof.  $\square$

**Lemma 4.12.** *We have*

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq c_N m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)).$$

*Proof.* Denote for brevity  $A = \Phi^{-1}(\text{supp}(f))$  and  $B = \Phi^{-1}(\text{supp}(1-\phi))$ . For  $\epsilon > 0$ , denote by  $A_\epsilon$  (respectively,  $B_\epsilon$ ) the  $\epsilon$ -neighborhood of  $A$  (respectively, of  $B$ ) with respect to the norm  $|\cdot|_\infty$ .

If  $2^{-n} < \epsilon$ , then the condition  $I \cap \text{supp}(\phi) \neq \emptyset$  yields, for  $Q = \Phi(I) \in \mathbb{D}$ , the condition  $Q \subset A_\epsilon$ . If  $2^{-n} < \frac{\epsilon}{3}$ , then the condition  $I \cap \text{supp}(1-\phi) \neq \emptyset$  yields, for  $Q = \Phi(I) \in \mathbb{D}$ , the condition  $Q \subset B_\epsilon$ . Hence, for  $2^{-n} < \frac{\epsilon}{3}$ , we have

$$\left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq \det(\Phi)^{-\frac{1}{N}} \left\| \left\{ m(Q)^{\frac{1}{N}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset A_\epsilon \cap B_\epsilon \\ \ell(Q) \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D})}.$$

Using Lemma 4.11 and passing  $n \rightarrow \infty$ , we obtain

$$\lim_{n \rightarrow \infty} \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq c_N \det(\Phi)^{-\frac{1}{N}} m^{\frac{1}{N}}(A_\epsilon \cap B_\epsilon).$$

Since  $A$  and  $B$  are both closed, it follows that

$$\lim_{\epsilon \rightarrow 0} m(A_\epsilon \cap B_\epsilon) = m(A \cap B).$$

Passing  $\epsilon \rightarrow 0$  in the second last display, we complete the proof.  $\square$

**Lemma 4.13.** *Let  $r > 2$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) &\leq \\ &\leq c_N \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1-\phi)). \end{aligned}$$

*Proof.* For a given  $n \in \mathbb{Z}$ , we have

$$\left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq \|\nabla f\|_{L_r(\mathbb{R}^N)} \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}}.$$

The right hand side obviously belongs to  $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$  and, hence, so does the left hand side. Thus,

$$\begin{aligned} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) &\leq \\ &\leq \left\| \left\{ m(I)^{\frac{1}{N} - \frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)} \leq \\ &\leq \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Passing  $n \rightarrow \infty$  and using Lemma 4.12, we complete the proof.  $\square$

**Lemma 4.14.** *For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq$$

$$\leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1 - \phi)).$$

*Proof.* For a given  $n \in \mathbb{Z}$ , we have

$$\left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I > 2^{-n}}} \leq c'_\Phi \|f\|_{L_1(\mathbb{R}^N)} \left\{ m(I)^{-1} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}.$$

The right hand side obviously belongs to  $(l_{N,\infty})_0(\mathbb{D}_\Gamma)$  and, hence, so does the left hand side. Thus,

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq \left\| \left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Obviously,

$$|f(x) - f(y)| \leq c_N |x - y| \cdot \|\nabla f\|_{L_\infty(\mathbb{R}^N)}.$$

Hence,

$$\begin{aligned} \|f - f \circ g\|_{L_1(I)} & \leq c_N \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \int_I |x - gx| dx \leq \\ & \leq c''_\Phi m(I) \times |c_I - gc_I| \times \|\nabla f\|_{L_\infty(\mathbb{R}^N)}. \end{aligned}$$

Thus,

$$\frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \leq c'''_\Phi m(I)^{\frac{1}{N}} \|\nabla f\|_{L_\infty(\mathbb{R}^N)}$$

Finally,

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{|c_I - gc_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times \left\| \left\{ m(I)^{\frac{1}{N}} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset \\ d_I \leq 2^{-n}}} \right\|_{l_{N,\infty}(\mathbb{D}_\Gamma)}. \end{aligned}$$

Passing  $n \rightarrow \infty$  and using Lemma 4.12, we complete the proof.  $\square$

**Lemma 4.15.** *Let  $r > 2$ . For every  $f \in C_c^\infty(\mathbb{R}^N)$  and for every  $n \in \mathbb{N}$ , we have*

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - gc_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq c_\Phi \|\nabla f\|_{L_\infty(\mathbb{R}^N)} \times m^{\frac{1}{N}}(\text{supp}(f) \cap \text{supp}(1 - \phi)). \end{aligned}$$

*Proof.* We obviously have

$$\begin{aligned} \|f - f \circ g\|_{L_r(I)} & \leq \left\| f - \frac{1}{m(I)} \int_I f \right\|_{L_r(I)} + \\ & + \left\| f \circ g - \frac{1}{m(I)} \int_I (f \circ g) \right\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)}. \end{aligned}$$

By Poincare inequality,

$$\|f - f \circ g\|_{L_r(I)} \leq$$

$$\leq c'_{\Phi,r} \left( m(I)^{\frac{1}{N}} \|\nabla f\|_{L_r(I)} + m(I)^{\frac{1}{N}} \|\nabla(f \circ g)\|_{L_r(I)} + m(I)^{\frac{1}{r}-1} \|f - f \circ g\|_{L_1(I)} \right).$$

By the triangle inequality,

$$\begin{aligned} & \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-\frac{1}{r}}}{|c_I - g c_I|} \|f - f \circ g\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) \leq \\ & \leq c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ & + c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ m(I)^{\frac{1}{N}-\frac{1}{r}} \|\nabla f\|_{L_r(gI)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right) + \\ & + c''_{\Phi,r} \text{dist}_{l_{N,\infty}(\mathbb{D}_\Gamma)} \left( \left\{ \frac{d_I m(I)^{-1}}{|c_I - g c_I|} \|f - f \circ g\|_{L_1(I)} \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ I \cap \text{supp}(f) \neq \emptyset \\ 3I \cap \text{supp}(1-\phi) \neq \emptyset}}, (l_{N,\infty})_0(\mathbb{D}_\Gamma) \right). \end{aligned}$$

The assertion follows from Lemmas 4.13 and 4.14.  $\square$

*Proof of Theorem 4.2.* The assertion follows by combining Lemmas 4.10 (applied with  $f - f \circ g$  instead of  $f$ ) and 4.15.  $\square$

## 5. SECOND ROCHBERG-SEMMES BASED ESTIMATE

**Theorem 5.1.** *Let  $N > 1$ . Fix a closed Weyl chamber  $\Gamma$ . Suppose that  $K : \Gamma \times \Gamma \rightarrow \mathbb{C}$  satisfies*

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K,\alpha,\beta} \frac{|x - y|^{-|\alpha|_1 - |\beta|_1}}{V(x, |x - y|)}, \quad x, y \in \Gamma, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

*Let  $V$  be an integral operator with integral kernel  $K$ . For every  $f \in \dot{W}^{1,N}(\Gamma)$ , we have*

$$\|[M_f, V]\|_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \leq c_{K,\kappa} \|f\|_{\dot{W}^{1,N}(\Gamma)}.$$

**Theorem 5.2.** *In the setting of Theorem 5.1, let  $f \in C_c^\infty(\mathbb{R}^N)$ . If  $0 \leq \phi_m \leq 1$  for  $m \geq 1$  are in  $C_c^\infty(\mathbb{R}^N)$  and  $\text{supp}(1 - \phi_m)$  increases to  $\mathbb{R}^N$ , then*

$$\text{dist}_{\mathcal{L}_{N,\infty}(L_2(\Gamma, w(x)dx))} \left( M_{\phi_m} \cdot [M_f, V] \cdot M_{\phi_m} - [M_f, V], (\mathcal{L}_{N,\infty})_0(L_2(\Gamma, w(x)dx)) \right) \rightarrow 0$$

as  $m \rightarrow \infty$ .

**Lemma 5.3.** *Assume that  $N > 1$  and  $2 < r < \infty$ . Let  $f \in L_r^{\text{loc}}(\mathbb{R}^N)$ . Fix a closed Weyl chamber  $\Gamma$ . Let  $K$  be as in Theorem 4.1. We have*

$$\begin{aligned} (f(x) - f(y)) \cdot K(x, y) &= \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k,l,I}^{(1)}(x) f_{k,l,I}^{(1)}(y) - \\ &- \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{I \in \mathbb{D}_\Gamma} e_{k,l,I}^{(2)}(x) f_{k,l,I}^{(2)}(y) \end{aligned}$$

where

- (i)  $e_{k,l,I}^{(1)}$  and  $e_{k,l,I}^{(2)}$  are supported in  $I$  and  $f_{k,l,I}^{(1)}$  and  $f_{k,l,I}^{(2)}$  are supported in  $3I$ ;

(ii) for every  $k, l \in \mathbb{Z}^N$  and for every  $I \in \mathbb{D}_\Gamma$ , we have

$$\begin{aligned} \|e_{k,l,I}^{(1)}\|_{L_r(\mathbb{R}^N, w(x)dx)} &\leq \left\| f - \frac{1}{m(3I)} \int_{3I} f \right\|_{L_r(3I, w(x)dx)}; \\ \|f_{k,l,I}^{(2)}\|_{L_r(\mathbb{R}^N, w(x)dx)} &= \left\| f - \frac{1}{m(3I)} \int_{3I} f \right\|_{L_r(3I, w(x)dx)}; \end{aligned}$$

(iii) for every  $k, l \in \mathbb{Z}^N$  and for every  $I \in \mathbb{D}_\Gamma$ , we have

$$\|f_{k,l,I}^{(1)}\|_{L_\infty(\mathbb{R}^N)}, \|e_{k,l,I}^{(2)}\|_{L_\infty(\mathbb{R}^N)} \leq c_{K,\kappa,\Phi};$$

*Proof.* Repeating the argument in Lemma 4.5 *mutatis mutandi*, we obtain

$$(1 + |k|^2 + |l|^2)^{-N-1} \cdot |\text{coef}_{k,l,I}| \leq \frac{c_{K,\kappa,\Phi}}{w(I)}, \quad k, l \in \mathbb{Z}^N, \quad I \in \mathbb{D}_\Gamma.$$

Set

$$\begin{aligned} e_{k,l,I}^{(1)}(x) &= \left( f(x) - \frac{1}{m(3I)} \int_{3I} f \right) \cdot \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x-c_I)}{d_I} \rangle}, \\ f_{k,l,I}^{(1)}(y) &= (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,I} \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y-c_I)}{d_I} \rangle}, \\ e_{k,l,I}^{(2)}(x) &= (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,I} \cdot \chi_I(x) e^{i\langle k, \frac{\Phi^{-1}(x-c_I)}{d_I} \rangle}, \\ f_{k,l,I}^{(2)}(y) &= \left( f(y) - \frac{1}{m(3I)} \int_{3I} f \right) \cdot \chi_{3I}(y) e^{i\langle l, \frac{\Phi^{-1}(y-c_I)}{d_I} \rangle}. \end{aligned}$$

The assertion follows immediately.  $\square$

to be written by Zhijie

## 6. PROOF OF THEOREM 1.1 (i),(iii)

Denote for brevity

$$V(x, y, t) = \max\{w(B(x, t)), w(B(y, t))\}$$

**Lemma 6.1.** For every pair of multi-indices  $\alpha, \beta \in \mathbb{Z}_+^N$ , for every  $t > 0$ , we have

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa,\alpha,\beta} t^{1 - \frac{|\alpha|_1 + |\beta|_1}{2}} |x - y|^{-2} V(x, y, t^{\frac{1}{2}})^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

*Proof.* This is (slightly weakened form of) Theorem 3.1 in [2].  $\square$

**Lemma 6.2.** Suppose  $s + N > 2$ . We have

$$\int_0^\infty t^{-\frac{s}{2}} V(x, y, t^{\frac{1}{2}})^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa,s} V(x, y, d(x, y))^{-1} \times d(x, y)^{2-s}.$$

*Proof.* By (2),

$$V(x, y, t^{\frac{1}{2}})^{-1} \leq c_\kappa^{(1)} V(x, y, d(x, y))^{-1} \times \begin{cases} \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} &\int_0^\infty t^{-\frac{s}{2}} V(x, y, t^{\frac{1}{2}})^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq \\ &\leq c_\kappa^{(1)} V(x, y, d(x, y))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left( \frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt + \end{aligned}$$

$$\begin{aligned}
& +c_{\kappa}^{(1)}V(x,y,d(x,y))^{-1} \times \int_{d(x,y)^2}^{\infty} t^{-\frac{s}{2}} \left( \frac{d(x,y)}{t^{\frac{1}{2}}} \right)^N \exp\left(-\frac{c_{\kappa}}{t}d(x,y)^2\right)dt = \\
& = c_{\kappa}^{(1)}V(x,y,d(x,y))^{-1} \times d(x,y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp\left(-\frac{c_{\kappa}}{t}\right)dt + \\
& + c_{\kappa}^{(1)}V(x,y,d(x,y))^{-1} \times d(x,y)^{2-s} \times \int_1^{\infty} t^{-\frac{s+N}{2}} \exp\left(-\frac{c_{\kappa}}{t}\right)dt.
\end{aligned}$$

□

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

**Lemma 6.3.** *Let  $N \geq 2$ . For every pair of multi-indices  $\alpha, \beta$  and for every  $1 \leq j \leq N$ , we have*

$$|(\partial_x^{\alpha} \partial_y^{\beta} K_{R_{\text{Dunkl},j}})(x,y)| \leq c_{\kappa,\alpha,\beta} \frac{d(x,y)^{1-|\alpha|_1-|\beta|_1}}{|x-y| \cdot V(x,y,d(x,y))}, \quad x,y \in \mathbb{R}^N.$$

*Proof.* We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^{\infty} e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x,y) = \pi^{-\frac{1}{2}} \int_0^{\infty} h_t(x,y) \frac{dt}{t^{\frac{1}{2}}}.$$

Obviously,

$$K_{R_{\text{Dunkl},j}}(x,y) = (T_j K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x,y),$$

where  $T_j$  on the right hand side acts in the  $x$  variable. Taking into account that

$$(T_j h_t)(x,y) = \frac{(y-x)_j}{2t} h_t(x,y),$$

we conclude that

$$K_{R_{\text{Dunkl},j}}(x,y) = \frac{(y-x)_j}{2\pi^{\frac{1}{2}}} \int_0^{\infty} h_t(x,y) \frac{dt}{t^{\frac{3}{2}}}.$$

By the Leibniz rule,

$$\begin{aligned}
(\partial_x^{\alpha} \partial_y^{\beta} K_{R_{\text{Dunkl},j}})(x,y) &= \frac{(y-x)_j}{2\pi^{\frac{1}{2}}} \int_0^{\infty} (\partial_x^{\alpha} \partial_y^{\beta} h_t)(x,y) \frac{dt}{t^{\frac{3}{2}}} - \\
&- \frac{\alpha_j}{2\pi^{\frac{1}{2}}} \int_0^{\infty} (\partial_x^{\alpha-e_j} \partial_y^{\beta} h_t)(x,y) \frac{dt}{t^{\frac{3}{2}}} + \frac{\beta_j}{2\pi^{\frac{1}{2}}} \int_0^{\infty} (\partial_x^{\alpha} \partial_y^{\beta-e_j} h_t)(x,y) \frac{dt}{t^{\frac{3}{2}}}.
\end{aligned}$$

Of course, if  $\alpha_j = 0$  (respectively,  $\beta_j = 0$ ), then the second (respectively, the third) summand on the right hand side does not exist.

Using Lemma 6.1, we estimate

$$\begin{aligned}
& |(\partial_x^{\alpha} \partial_y^{\beta} K_{R_{\text{Dunkl},j}})(x,y)| \leq \\
& \leq \frac{c_{\kappa,\alpha,\beta}}{2\pi^{\frac{1}{2}}|x-y|} \int_0^{\infty} t^{-\frac{1+|\alpha|_1+|\beta|_1}{2}} V(x,y,t^{\frac{1}{2}})^{-1} \exp\left(-\frac{c_{\kappa}}{t}d(x,y)^2\right)dt + \\
& + \frac{\alpha_j c_{\kappa,\alpha-e_j,\beta}}{2\pi^{\frac{1}{2}}|x-y|^2} \int_0^{\infty} t^{-\frac{|\alpha|_1+|\beta|_1}{2}} V(x,y,t^{\frac{1}{2}})^{-1} \exp\left(-\frac{c_{\kappa}}{t}d(x,y)^2\right)dt + \\
& + \frac{\beta_j c_{\kappa,\alpha,\beta-e_j}}{2\pi^{\frac{1}{2}}|x-y|^2} \int_0^{\infty} t^{-\frac{|\alpha|_1+|\beta|_1}{2}} V(x,y,t^{\frac{1}{2}})^{-1} \exp\left(-\frac{c_{\kappa}}{t}d(x,y)^2\right)dt.
\end{aligned}$$

Applying Lemma 6.2, we conclude

$$\begin{aligned}
& |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| \leq \\
& \leq \frac{c_{\kappa, \alpha, \beta}}{2\pi^{\frac{1}{2}} |x - y|} \times c_{\kappa, 1+|\alpha|_1+|\beta|_1} V(x, y, d(x, y))^{-1} \times d(x, y)^{1-|\alpha|_1-|\beta|_1} + \\
& + \frac{\alpha_j c_{\kappa, \alpha - e_j, \beta}}{2\pi^{\frac{1}{2}} |x - y|^2} \times c_{\kappa, |\alpha|_1+|\beta|_1} V(x, y, d(x, y))^{-1} \times d(x, y)^{2-|\alpha|_1-|\beta|_1} + \\
& + \frac{\beta_j c_{\kappa, \alpha, \beta - e_j}}{2\pi^{\frac{1}{2}} |x - y|^2} \times c_{\kappa, 1+|\alpha|_1+|\beta|_1} V(x, y, d(x, y))^{-1} \times d(x, y)^{1-|\alpha|_1-|\beta|_1}.
\end{aligned}$$

The assertion follows immediately.  $\square$

**Lemma 6.4.** *Let  $\Gamma$  be an open chamber and let  $1 \neq g \in G$ . We have*

- (i)  $\bar{\Gamma} \cap g\bar{\Gamma}$  is  $k$ -dimensional chamber of  $\Gamma$  for some  $1 \leq k < d$ .
- (ii)  $g(x) = x$  for  $x \in \bar{\Gamma} \cap g\bar{\Gamma}$ .

*Proof.* must be a standard material. reference to be found.  $\square$

**Lemma 6.5.** *Let  $N \geq 2$ ,  $1 \leq N_1 \leq N$  and  $N_2 = N - N_1$ . Let  $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . Let  $F$  be a face of  $\Omega$  (not necessarily of maximal dimension, possibly of zero dimension). Let  $A : \mathbb{R}^N \rightarrow \mathbb{R}^N$  be a bijective linear map. Suppose  $A(\bar{\Omega}) \cap \bar{\Omega} = F$ ,  $A|_F = \text{id}$ . Suppose also that  $A : \mathbb{R}^{N_1} \times \{0\} \rightarrow \mathbb{R}^{N_1} \times \{0\}$ ,  $A : \{0\} \times \mathbb{R}^{N_2} \rightarrow \{0\} \times \mathbb{R}^{N_2}$  and  $A|_{\{0\} \times \mathbb{R}^{N_2}} = \text{id}$ . Let  $\sigma : \mathbb{R}^N \times \mathbb{R}^N$  be the reflection with respect to  $F$ . We have*

$$|x - Ay| \geq c_A |x - \sigma y|, \quad x, y \in \Omega.$$

*Proof.* In what follows, we prove the assertion for the case when  $F$  has non-zero dimension. The case  $\dim(F) = 0$  (i.e., the case  $F = \{0\}$ ) is similar but simpler.

Since  $A, \sigma : \mathbb{R}^{N_1} \times \{0\} \rightarrow \mathbb{R}^{N_1} \times \{0\}$ ,  $A, \sigma : \{0\} \times \mathbb{R}^{N_2} \rightarrow \{0\} \times \mathbb{R}^{N_2}$ , and  $A|_{\{0\} \times \mathbb{R}^{N_2}} = \text{id}$  and  $\sigma|_{\{0\} \times \mathbb{R}^{N_2}} = \text{id}$ , it suffices to prove the assertion for the case  $N_1 = N$ ,  $N_2 = 0$ .

Without loss of generality, we may assume that  $F = \mathbb{R}_+^k \times \{0\}$  with  $1 \leq k \leq N$ .

Since  $A|_{\mathbb{R}^k \times \{0\}} = \text{id}$ , it follows that

$$A = \begin{pmatrix} 1 & B \\ 0 & D \end{pmatrix}$$

where  $1$  is  $k \times k$  matrix,  $B$  is  $k \times (N - k)$  matrix and  $D$  is  $(N - k) \times (N - k)$  matrix.

**Step 1:** We claim that  $D(\mathbb{R}_+^{N-k}) \cap \mathbb{R}_+^{N-k} = \{0\}$ .

Let  $w \in \mathbb{R}_+^{N-k}$  be such that  $Dw \geq 0$ . Set  $x = (t, \dots, t, w) \in \mathbb{R}^N$ , where  $t \in \mathbb{R}_+$  is such that  $t \geq |Bw|_\infty$ . We have  $x \geq 0$  (this is obvious) and  $Ax \geq 0$  (because  $t(1, \dots, 1) + Bw \geq 0$  and  $Dw \geq 0$ ). Since  $A(\mathbb{R}_+^N) \cap \mathbb{R}_+^N = \mathbb{R}_+^k \times \{0\}$ , it follows that  $x \in \mathbb{R}_+^k \times \{0\}$ . In other words,  $w = 0$ .

**Step 2:** It follows from Step 1 that  $z - Dw \neq 0$  for  $z, w \in \mathbb{R}_+^{N-k}$  (unless  $z = w = 0$ ). Hence,  $|z - Dw| \geq c_D(|z| + |w|)$  for  $z, w \in \mathbb{R}_+^{N-k}$ .

Set  $x = (u, z)$  and  $y = (v, w)$  with  $u, v \in \mathbb{R}_+^k$  and  $z, w \in \mathbb{R}_+^{N-k}$ . We have

$$|x - Ay| = |(u - v - Bw, z - Dw)| \geq |z - Dw| \geq c_D(|z| + |w|).$$

On the other hand, we have

$$|x - Ay| = |(u - v - Bw, z - Dw)| \geq |u - v| - |Bw| \geq |u - v| - \|B\|_\infty |w|.$$

Thus,

$$(1 + c_D^{-1} \|B\|_\infty) |x - Ay| \geq |u - v|.$$

Finally,

$$(1 + c_D^{-1} \|B\|_\infty + c_D^{-1})|x - Ay| \geq |u - v| + |z| + |w| \geq |(u - v, z + w)| = |x - \sigma y|.$$

This completes the proof.  $\square$

*Proof of Theorem 1.1 (iii).* We prove the assertion under *a priori* restriction  $f \in C_c^\infty(\mathbb{R}^N)$ . Passing to the case of  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$  is standard (see e.g. [refer to our papers](#)).

It suffices to estimate, for every chamber  $\Gamma$  and for every  $g \in G$ , the operator  $M_{\chi_\Gamma}[R_{\text{Dunkl},j}, M_f]M_{\chi_\Gamma}$ . In other words, it suffices to estimate an operator with integral kernel

$$(f(y) - f(x))K_{R_{\text{Dunkl},j}}(x, y), \quad x \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space  $L_2(\mathbb{R}^N, w(x)dx)$ . In other words, it suffices to estimate an operator with integral kernel

$$(f(gy) - f(x))K_{R_{\text{Dunkl},j}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space  $L_2(\Gamma, w(x)dx)$ .

Let  $N_1 = \dim(\text{span}(R))$ ,  $N_2 = N - N_1$  and let  $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ . For a concrete chamber  $\Gamma$ , let  $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$  be a linear isomorphism given in Theorem 2.2 so that  $\Phi(\Gamma) = \Omega$ . Without loss of generality,  $\Phi(\text{span}(R)) = \mathbb{R}^{N_1} \times \{0\}$  and  $\Phi(\text{span}(R)^\perp) = \{0\} \times \mathbb{R}^{N_2}$ .

It suffices to estimate an operator with integral kernel

$$(f(g\Phi^{-1}y) - f(\Phi^{-1}x))K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w(\Phi^{-1}x)dx)$ . Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$(f(g\Phi^{-1}y) - f(\Phi^{-1}x))K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$ .

For  $g = 1_G$ , we need to estimate an operator with integral kernel

$$(f(\Phi^{-1}y) - f(\Phi^{-1}x))K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, \Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$ .

For  $g \in 1_G$ , it suffices to estimate operators with integral kernels

$$(f(g\Phi^{-1}y) - f(g\Phi^{-1}x))K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

and, respectively,

$$(f(g\Phi^{-1}x) - f(\Phi^{-1}x))K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$ .

Denote by  $V_{g,G,j}$  integral operator on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$  with integral kernel

$$K_{g,G,j}(x, y) = K_{R_{\text{Dunkl},j}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega.$$

For  $g = 1_G$ , it suffices to estimate operator  $[M_{f \circ \Phi^{-1}}, V_{g,G,j}]$  on the Hilbert space  $L_2(\Omega, w_\theta(x)dx)$ . By Lemmas 6.3 and 3.5, for every pair of multi-indices  $\alpha, \beta$ , we have

$$|(\partial_x^\alpha \partial_y^\beta K_{g,G,j})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{|x - y|^{-|\alpha|_1 - |\beta|_1}}{w_\theta(x, |x - y|)}, \quad x, y \in \Omega.$$

Applying Theorem 5.1 to the operator  $V_{g,G,j}$  and to the function  $f \circ \Phi^{-1}$ , we obtain the required estimate.

For  $g \neq 1_G$ , it suffices to estimate operators  $[M_{f \circ g \Phi^{-1}}, V_{g,G,j}]$  and  $M_{f \circ g \Phi^{-1} - f \circ \Phi^{-1}} V_{g,G,j}$ . By Lemmas 6.3 and 3.5, for every pair of multi-indices  $\alpha, \beta$  and for every  $1 \leq j \leq N$ , we have

$$|(\partial_x^\alpha \partial_y^\beta K_{g,G,j})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{|x - y|}{|x - \Phi g \Phi^{-1} y|} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(x, |x - y|)}, \quad x, y \in \Omega.$$

By Lemma 6.4, the matrix  $\Phi g \Phi^{-1}$  satisfies the assumptions in Lemma 6.5. Let  $F$  be a face of  $\Omega$  (not necessarily of maximal dimension) and let  $\sigma$  be the reflection with respect to  $F$  as given by Lemma 6.5. It follows from Lemma 6.5 that

$$|(\partial_x^\alpha \partial_y^\beta K_{g,G,j})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{|x - y|}{|x - \sigma y|} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(x, |x - y|)}, \quad x, y \in \Omega.$$

Applying Theorem 5.1 to the operator  $V_{g,G,j}$  and to the function  $f \circ g \Phi^{-1}$ , we obtain the required estimate for  $[M_{f \circ g \Phi^{-1}}, V_{g,G,j}]$ . Applying Theorem 4.1 to the operator  $V_{g,G,j}$  and to the function  $f \circ g \Phi^{-1} - f \circ \Phi^{-1}$ , we obtain the required estimate for  $M_{f \circ g \Phi^{-1} - f \circ \Phi^{-1}} V_{g,G,j}$ .

This completes the proof.  $\square$

## 7. PROOF OF THEOREM 1.1 (ii), (iv)

**Lemma 7.1.** *Let  $Q \in \mathcal{D}$ . We have*

$$K_j(x, y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

*Proof.* **TO BE DONE. Follows from Lemma ??.**  $\square$

**Lemma 7.2.** *Let real-valued  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$  be such that the commutator  $[R_{\text{Dunkl}, j}, M_f]$  is bounded on  $L_2(\mathbb{R}^N, w(x)dx)$ . For every  $Q \in \mathcal{D}$ , there exist  $\xi_Q$  and  $\eta_Q$  supported in  $7Q$  such that  $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$  and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_k |\langle [R_{\text{Dunkl}, j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

*Proof.* Let  $\alpha$  be the median of  $f$  on  $3Q$  and  $\beta$  be the median of  $f$  on  $Q + 3s(Q)e_j$ . Assume for definiteness  $\alpha \geq \beta$ . Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, & B_1 &= \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, & B_2 &= \{x \in Q + 3s(Q)e_j : f(x) \geq \alpha\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, & \eta_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} &\langle [R_{\text{Dunkl}, j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ &= \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If  $x \in B_1$  and  $y \in A_1$ , then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If  $x \in B_2$  and  $y \in A_2$ , then

$$f(y) - f(x) = (f(y) - \beta) + (\beta - f(x)) =$$

$$= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0.$$

Thus,

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ & \geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ & \geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ & = \frac{c_k}{w^2(Q)} \cdot \left( w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ & = \frac{c_k}{w^2(Q)} \cdot \left( \frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ & \geq \frac{c_k}{4w(Q)} \cdot \left( \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} & -\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ & \geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ & \geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ & = \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ & = \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} & \int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ & + \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ & \leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ & + \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ & \leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$ ,  
then we set  $\xi_Q = \xi_{Q,1}$  and  $\eta_Q = \eta_{Q,1}$ . Otherwise, set  $\xi_Q = \xi_{Q,2}$  and  $\eta_Q = \eta_{Q,2}$ . Thus,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c'_k |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

**Lemma 7.3.** *Suppose that  $1 < p < \infty$  and  $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ . If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))$  for some  $j \in \{1, 2, \dots, N\}$ , then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathcal{D}} \right\|_{l_p(\mathcal{D})} \leq \\ & \leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

If  $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$  for some  $j \in \{1, 2, \dots, N\}$ , then

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathcal{D}} \right\|_{l_{N,\infty}(\mathcal{D})} \leq \\ & \leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

*Proof.* We prove only the first assertion. The proof of the second one follows *mutatis mutandi*.

Let  $\{\xi_Q\}_{Q \in \mathcal{D}}$  and  $\{\eta_Q\}_{Q \in \mathcal{D}}$  be functions constructed in Lemma 7.2. By Lemma ??,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathcal{D}} \in l_p.$$

The first assertion follows now from Lemma 7.2. □

**Lemma 7.4.** *Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_k}{m(Q)} \int_Q w(x) dx.$$

*Proof.* If  $x = c_Q + y$ ,  $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$ , then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber  $\Gamma$  such that  $c_Q \in \Gamma$ . We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If  $y \in \Gamma$ , then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x) dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone  $\Gamma' \subset \Gamma$  such that (i) interior of  $\Gamma'$  is non-empty and (ii)  $\partial\Gamma \cap \partial\Gamma' = \{0\}$ . We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x) dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\frac{1}{4}s(Q) \leq |y| \leq \frac{1}{2}s(Q)} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2} c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4} c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof.  $\square$

**Lemma 7.5.** *Suppose  $1 < t < 1 + \frac{1}{|R| \max_{v \in R} k(v)}$ . Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). We have*

$$\left( \frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} \leq c_{t,k} \frac{m(Q)}{\int_Q w(x) dx}.$$

*Proof.* If  $v \in R$  is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2} s(Q) \geq \frac{1}{2} |\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq 2^{(t-1)|R|k(v)} |\langle c_Q, v \rangle|^{(1-t)|R|k(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|k(v)} |\langle x_0, v \rangle|^{(1-t)|R|k(v)} \cdot s(Q)^N. \end{aligned}$$

If  $v \in R$  is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let  $y_0 = \langle c_Q, v \rangle$  and  $z_0 = c_Q - y_0 v$ . Let us introduce new variables  $y = \langle x, v \rangle$  and  $z = x - yv$  (it varies in the plane orthogonal to  $v$ ). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2} s(Q)} |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2} r} |y|^{(1-t)|R|k(v)} dy dz = \\ &= \left( \frac{\sqrt{d}}{2} s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2} s(Q)} |y|^{(1-t)|R|k(v)} dy \leq \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{\sqrt{d}}{2}s(Q)\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy = \\
&= s(Q)^{N+(1-t)|R|k(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|k(v)} dy.
\end{aligned}$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\begin{aligned}
\int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left( \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \right)^{\frac{1}{|R|}} \leq \\
&\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)k(v)}.
\end{aligned}$$

The assertion follows easily.  $\square$

**Lemma 7.6.** *Measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [8].*

*Proof.* That both measures are doubling is obvious.

Let  $Q$  be a cube in  $\mathbb{R}^N$  (with sides parallel to coordinate hyperplanes). By Lemmas 7.4 and 7.5,

$$\begin{aligned}
\left(\frac{1}{m(Q)} \int_Q w^t(x) dx\right)^{\frac{1}{t}} &\leq \frac{c_k}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\
\left(\frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx}\right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}.
\end{aligned}$$

The assertion follows now from Proposition 3.6 in [8].  $\square$

*Proof of Theorem 1.1 (ii).* The left hand side in Lemma 7.3 is the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . So, the oscillation semi-norm of the function  $f$  with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl}, j}, M_f]$ .

Lemma 7.6 asserts that measures  $dx$  and  $w(x)dx$  (and distance  $(x, y) \rightarrow |x - y|_\infty$ ) satisfy the conditions in Proposition 1.2 in [8]. Appealing to Proposition 1.2 in [8], we conclude that the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the one with respect to measure  $w(x)dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$ . By the preceding paragraph, the oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is controlled by the  $\|\cdot\|_{\mathcal{L}_p(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator  $[R_{\text{Dunkl}, j}, M_f]$ .

That oscillation semi-norm of the function  $f$  with respect to measure  $dx$  and distance  $(x, y) \rightarrow |x - y|_\infty$  is equivalent to the Sobolev semi-norm of the function  $f$  is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [9]. The assertion follows now from the preceding paragraph.  $\square$

*Proof of Theorem 1.1 (iv).* The argument follows the one in the proof of Theorem 1.1 (ii) *mutatis mutandi*.  $\square$

## 8. SPECTRAL ASYMPTOTIC FORMULA

**Definition 8.1.** For  $g \in G$  and  $j = 1, \dots, N$ , set

$$d_{g,j} := g^{-1}e_j, \quad f_g(x) := f(gx).$$

For  $\alpha \in R_+$ , define  $S_\alpha : \mathbb{C}^G \rightarrow \mathbb{C}^G$  by

$$(S_\alpha z)_g := z_{g\sigma_\alpha}, \quad g \in G.$$

For  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,  $x \in \Gamma_+$ , and  $\theta \in \mathbb{S}^{N-1}$ , define

$$\begin{aligned} \mathbf{c}_{f,j}(x, \theta) &:= \text{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &\quad + \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

For smooth  $f$ , the quotients in the second term are understood by their smooth extensions across the corresponding reflection hyperplanes. For  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,  $\mathbf{c}_{f,j}$  is understood almost everywhere on  $\Gamma \times \mathbb{S}^{N-1}$ .

**Lemma 8.2.** Let  $f \in W_{\text{loc}}^{1,1}(\mathbb{R}^N)$ . For every  $u \in C_c^\infty(\mathbb{R}^N)$ ,

$$(1) \quad [T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x)$$

for almost every  $x \in \mathbb{R}^N$ . If  $N > 1$  and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ , then, for every  $\alpha \in R_+$ ,

$$(2) \quad \left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* Pairing  $\alpha$  and  $-\alpha$  in the definition of  $T_{e_j}$ , and using  $k(-\alpha) = k(\alpha)$  and  $\sigma_{-\alpha} = \sigma_\alpha$ , we have

$$T_{e_j}u(x) = \partial_j u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{u(x) - u(\sigma_\alpha x)}{\langle \alpha, x \rangle}.$$

Therefore,

$$[T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x),$$

which proves (1).

Since  $R$  is normalized,

$$x - \sigma_\alpha x = \langle \alpha, x \rangle \alpha, \quad \|x - \sigma_\alpha x\| = \sqrt{2} |\langle \alpha, x \rangle|.$$

Hence, for almost every  $x \notin \alpha^\perp$ ,

$$\frac{|f(x) - f(\sigma_\alpha x)|}{|\langle \alpha, x \rangle|} = \sqrt{2} \mathcal{G}_{I, \sigma_\alpha} f(x).$$

By Lemma ?? (??),

$$\left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

This completes the proof of Lemma 8.2.  $\square$

**Lemma 8.3.** *Let  $\Gamma_+$  be an open Weyl chamber, let  $\Gamma$  be its closure, and let  $R_+$  be the corresponding positive subsystem. Set*

$$\rho(x) := \varpi(x)^{1/2},$$

and define

$$\begin{aligned} U : L^2(\mathbb{R}^N, dw) &\longrightarrow L^2(\Gamma, dw; \mathbb{C}^G), & (Uu)_g(x) &:= u(gx), \\ V : L^2(\Gamma, dw; \mathbb{C}^G) &\longrightarrow L^2(\Gamma, dx; \mathbb{C}^G), & (Vh)(x) &:= \rho(x)h(x). \end{aligned}$$

Then  $U$  and  $V$  are unitary. If

$$\tilde{T}_j := VUT_{e_j}U^{-1}V^{-1}, \quad \tilde{P} := VU(-\Delta_{\text{Dunkl}})U^{-1}V^{-1},$$

then, on  $C_c^\infty(\Gamma_+; \mathbb{C}^G)$ ,

$$(3) \quad (\tilde{T}_j h)_g = \partial_{d_{g,j}} h_g - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} h_{g\sigma_\alpha},$$

$$(4) \quad \tilde{P} = -\Delta_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Assume in addition that  $N > 1$ , and let

$$\tilde{Q} := VU(-\Delta_{\text{Dunkl}})^{-1/2}U^{-1}V^{-1},$$

where  $(-\Delta_{\text{Dunkl}})^{-1/2}$  is defined by the spectral calculus. Then, for every  $\chi \in C_c^\infty(\Gamma_+)$ , the operator  $M_\chi \tilde{Q} M_\chi$ , initially defined on its natural domain, admits a bounded extension to  $L^2(\Gamma, dx; \mathbb{C}^G)$ , and this extension is a classical matrix-valued pseudodifferential operator of order  $-1$ . Its principal symbol is

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

Consequently, if  $x \in \Gamma_+$ ,  $\xi \neq 0$ , and  $\chi \in C_c^\infty(\Gamma_+)$  is identically 1 in a neighborhood of  $x$ , then

$$(5) \quad \sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}.$$

*Proof.* The Weyl chambers  $g\Gamma_+$ ,  $g \in G$ , are pairwise disjoint and cover  $\mathbb{R}^N$  up to the reflection hyperplanes; see [6, Theorem 2.12]. Since  $dw$  is  $G$ -invariant,

$$\begin{aligned} \|Uu\|_{L^2(\Gamma, dw; \mathbb{C}^G)}^2 &= \sum_{g \in G} \int_{\Gamma} |u(gx)|^2 dw(x) \\ &= \int_{\mathbb{R}^N} |u(x)|^2 dw(x). \end{aligned}$$

Conversely, for  $h = (h_g)_{g \in G} \in L^2(\Gamma, dw; \mathbb{C}^G)$ , define  $u(gx) := h_g(x)$  for  $x \in \Gamma_+$ . This determines  $u$  almost everywhere on  $\mathbb{R}^N$  and gives  $Uu = h$ . Hence  $U$  is unitary. Since  $dw = \rho^2 dx$  and  $\rho > 0$  on  $\Gamma_+$ ,  $V$  is unitary as well.

Pairing the roots  $\alpha$  and  $-\alpha$  in the definition of  $T_{e_j}$ , changing variables in the root system by  $g$ , and using the  $G$ -invariance of  $k$ , we obtain, for  $h \in C_c^\infty(\Gamma_+; \mathbb{C}^G)$ ,

$$(UT_{e_j}U^{-1}h)_g = \partial_{d_{g,j}} h_g + \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (h_g - h_{g\sigma_\alpha}).$$

On  $\Gamma_+$ ,

$$\nabla \log \rho(x) = \sum_{\alpha \in R_+} k(\alpha) \frac{\alpha}{\langle \alpha, x \rangle},$$

and therefore

$$\rho \partial_{d_{g,j}} \rho^{-1} = \partial_{d_{g,j}} - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle}.$$

Conjugation by  $V$  cancels the diagonal terms and gives (3).

Similarly, pairing  $\alpha$  and  $-\alpha$  in the formula for the Dunkl Laplacian and using  $\|\alpha\|^2 = 2$ , we have

$$\begin{aligned} (U(-\Delta_{\text{Dunkl}})U^{-1}h)_g &= -\Delta h_g - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{\partial_\alpha h_g}{\langle \alpha, x \rangle} \\ &\quad + 2 \sum_{\alpha \in R_+} k(\alpha) \frac{h_g - h_{g\sigma_\alpha}}{\langle \alpha, x \rangle^2}. \end{aligned}$$

Writing  $\phi = \log \rho$ ,

$$\rho(-\Delta - 2\nabla\phi \cdot \nabla)\rho^{-1} = -\Delta + |\nabla\phi|^2 + \Delta\phi,$$

while

$$\Delta\phi = - \sum_{\alpha \in R_+} k(\alpha) \frac{\|\alpha\|^2}{\langle \alpha, x \rangle^2} = -2 \sum_{\alpha \in R_+} \frac{k(\alpha)}{\langle \alpha, x \rangle^2}.$$

The latter term cancels the diagonal part of

$$2 \sum_{\alpha \in R_+} k(\alpha) \frac{I_{\mathbb{C}^G} - S_\alpha}{\langle \alpha, x \rangle^2},$$

and hence

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2},$$

which proves (4).

Assume now that  $N > 1$ , fix  $\chi \in C_c^\infty(\Gamma_+)$ , and set

$$K := \text{supp } \chi.$$

On  $\Gamma_+$ , write

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + W(x), \quad W(x) = |\nabla \log \rho(x)|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Thus  $W$  is a smooth Hermitian matrix-valued function on  $\Gamma_+$ .

Choose open sets

$$K \Subset \Omega_0 \Subset \Omega_1 \Subset \Gamma_+$$

and  $\eta \in C_c^\infty(\Gamma_+)$  such that

$$0 \leq \eta \leq 1, \quad \eta = 1 \quad \text{on a neighborhood of } \overline{\Omega_1}.$$

Choose a flat torus  $M$  containing  $\text{supp } \eta$  in one Euclidean coordinate chart. Since  $\eta$  is compactly supported in that chart, the differential operator  $M_\eta \tilde{P} M_\eta$  extends smoothly by zero to  $M$ . Define on  $L^2(M; \mathbb{C}^G)$

$$(6) \quad P_0 := M_\eta \tilde{P} M_\eta + M_{1-\eta}(-\Delta_M + 1)M_{1-\eta}.$$

It is a nonnegative self-adjoint elliptic differential operator of order 2. Indeed,

$$\langle P_0 u, u \rangle = \langle \tilde{P}(\eta u), \eta u \rangle + \|\nabla((1-\eta)u)\|_2^2 + \|(1-\eta)u\|_2^2 \geq 0,$$

and its principal symbol is

$$p_{0,2}(x, \xi) = (\eta(x)^2 + (1-\eta(x))^2)|\xi|^2 I_{\mathbb{C}^G},$$

which is elliptic since

$$\eta^2 + (1-\eta)^2 \geq \frac{1}{2}.$$

Moreover,

$$(7) \quad P_0 = \tilde{P} \quad \text{on } \Omega_1.$$

Let

$$H_t := e^{-t\tilde{P}}, \quad H_t^0 := e^{-tP_0}.$$

We first consider

$$B_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} H_t^0 dt.$$

Let  $\Pi_0$  be the orthogonal projection onto  $\ker P_0$ . Since  $P_0$  is elliptic on the closed manifold  $M$ ,  $\Pi_0$  is smoothing. Applying the complex powers calculus [?, Theorem 2 and p. 290] to  $P_0 + \Pi_0$ , we obtain

$$(P_0 + \Pi_0)^{-1/2} \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G)$$

with principal symbol  $p_{0,2}^{-1/2}$ .

Set

$$F(\lambda) := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} e^{-t\lambda} dt, \quad \lambda \geq 0.$$

Then  $B_0 = F(P_0)$ , and for  $\lambda > 0$ ,

$$F(\lambda) - \lambda^{-1/2} = -\frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} e^{-t\lambda} dt.$$

Hence

$$F(\lambda) - \lambda^{-1/2} = O(\lambda^{-m}) \quad (\lambda \rightarrow \infty)$$

for every  $m > 0$ . On  $\ker P_0$  the difference between  $B_0$  and  $(P_0 + \Pi_0)^{-1/2}$  is finite rank. It follows from elliptic spectral regularity that

$$B_0 - (P_0 + \Pi_0)^{-1/2} \in \Psi^{-\infty}(M; \mathbb{C}^G).$$

Consequently,

$$(8) \quad B_0 \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G), \quad \sigma_{-1}(B_0)(x, \xi) = p_{0,2}(x, \xi)^{-1/2}.$$

In particular, by (7),

$$(9) \quad \sigma_{-1}(B_0)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}, \quad x \in \Omega_1.$$

We next compare  $H_t$  and  $H_t^0$  in the interior. Choose  $\zeta \in C_c^\infty(\Omega_1)$  such that

$$\zeta = 1 \quad \text{on a neighborhood of } K.$$

Since  $P_0 = \tilde{P}$  on  $\Omega_1$ , Duhamel's formula gives

$$(10) \quad M_\chi(H_t - H_t^0)M_\chi = - \int_0^t M_\chi H_{t-s} [\tilde{P}, M_\zeta] H_s^0 M_\chi ds.$$

Indeed, for data supported in  $K$ ,

$$v(t) := M_\zeta H_t^0 M_\chi u$$

satisfies

$$(\partial_t + \tilde{P})v = [\tilde{P}, M_\zeta] H_t^0 M_\chi u, \quad v(0) = M_\chi u,$$

which yields (10).

Let  $h_t(x, y)$  be the Dunkl heat kernel. The integral kernel of  $H_t$ , with respect to Lebesgue measure on  $\Gamma$ , is the matrix

$$(11) \quad \tilde{h}_t(x, y)_{g, \ell} = \rho(x)\rho(y) h_t(gx, \ell y), \quad g, \ell \in G.$$

For  $x, y \in \Gamma$ ,

$$d(gx, \ell y) = d(x, y) = |x - y|.$$

Together with the mixed derivative estimate for the Dunkl heat kernel [2, Theorem 3.1], this implies that, whenever  $E, F \Subset \Gamma_+$  and  $\text{dist}(E, F) > 0$ , for every pair of multi-indices  $\alpha, \beta$  and every  $m > 0$ ,

$$(12) \quad \sup_{\substack{x \in E \\ y \in F}} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

The heat kernel of the smooth elliptic operator  $P_0$  satisfies the same off-diagonal estimate on  $M$ .

Since

$$\text{supp}[\tilde{P}, M_\zeta] \cap K = \emptyset,$$

the two heat kernels occurring in (10) are evaluated between compact sets separated by a positive distance. The standard Gaussian bounds underlying (12) therefore give, for every multi-indices  $\alpha, \beta$  and every  $m > 0$ ,

$$(13) \quad \sup_{x, y} \left\| \partial_x^\alpha \partial_y^\beta K_{M_\chi(H_t - H_t^0)M_\chi}(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

Hence

$$(14) \quad S_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} M_\chi(H_t - H_t^0)M_\chi dt \in \Psi^{-\infty}.$$

It remains to treat the large-time part. Let  $K_1 \Subset \Gamma_+$  be a compact set containing  $K$ . By (11), the heat-kernel derivative estimate cited above, and (2), for  $x, y \in K_1$  and  $t \geq 1$ ,

$$V(gx, \ell y, \sqrt{t})^{-1} \lesssim_{K_1} t^{-N/2}.$$

Since  $\rho$  and all its derivatives are bounded on  $K_1$ , it follows that, for every pair of multi-indices  $\alpha, \beta$ ,

$$(15) \quad \sup_{x, y \in K_1} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| \lesssim_{\alpha, \beta, K_1} t^{-N/2}, \quad t \geq 1.$$

Since  $N > 1$ ,

$$\int_1^\infty t^{-1/2} t^{-N/2} dt < \infty.$$

Thus all derivatives of

$$K_{S_\infty}(x, y) := \frac{\chi(x)\chi(y)}{\sqrt{\pi}} \int_1^\infty t^{-1/2} \tilde{h}_t(x, y) dt$$

converge locally uniformly, and therefore

$$(16) \quad S_\infty := \frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} M_\chi H_t M_\chi dt \in \Psi^{-\infty}.$$

For completeness, the preceding integrals give the canonical bounded localization of  $\tilde{P}^{-1/2}$ . Indeed, for  $\varepsilon > 0$ ,

$$(\tilde{P} + \varepsilon)^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\varepsilon t} H_t dt$$

by the spectral theorem. The estimates (13) and (15) imply that

$$M_\chi (\tilde{P} + \varepsilon)^{-1/2} M_\chi$$

converges in operator norm, as  $\varepsilon \downarrow 0$ , to

$$\frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} M_\chi H_t M_\chi dt.$$

On the natural domain of  $\tilde{P}^{-1/2}$ , this limit agrees with  $M_\chi \tilde{P}^{-1/2} M_\chi$ ; hence it is its bounded extension.

Combining (14), (16), and the definition of  $B_0$ , we obtain

$$M_\chi \tilde{Q} M_\chi = M_\chi B_0 M_\chi + S_0 + S_\infty.$$

By (8),  $M_\chi B_0 M_\chi$  is a classical matrix-valued pseudodifferential operator of order  $-1$ , while  $S_0 + S_\infty$  is smoothing. Therefore

$$M_\chi \tilde{Q} M_\chi \in \Psi_{\text{cl}}^{-1},$$

and, by (9),

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

If  $\chi = 1$  in a neighborhood of  $x$ , this reduces to

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G},$$

which proves (5). □

**Lemma 8.4.** *Let  $j \in \{1, \dots, N\}$  and  $f \in C_c^\infty(\mathbb{R}^N)$ , and set*

$$A_{f,j} := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

*Then, for every  $\chi \in C_c^\infty(\Gamma_+)$ ,*

$$M_\chi A_{f,j} M_\chi$$

*is a compactly supported classical matrix valued pseudodifferential operator of order  $-1$ , with principal symbol*

$$(17) \quad \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) = \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

*Proof.* Set

$$F(x) := \text{diag}_{g \in G} (f_g(x)).$$

Since  $VUM_fU^{-1}V^{-1} = M_F$  and  $R_{\text{Dunkl},j} = T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$ ,

$$(18) \quad A_{f,j} = [\tilde{T}_j, M_F] \tilde{Q} + \tilde{T}_j [\tilde{Q}, M_F].$$

By (3),

$$(19) \quad \begin{aligned} [\tilde{T}_j, M_F] &= \text{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g \rangle) \\ &+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g - f_{g\sigma_\alpha}) \right) S_\alpha. \end{aligned}$$

Fix  $\chi \in C_c^\infty(\Gamma_+)$ . Choose  $\eta, \psi \in C_c^\infty(\Gamma_+)$  such that  $\eta = 1$  on a neighborhood of  $\text{supp } \chi$  and  $\psi = 1$  on a neighborhood of  $\text{supp } \eta$ , and set

$$Q_\psi := M_\psi \tilde{Q} M_\psi.$$

By Lemma 8.3,  $Q_\psi$  is a classical matrix valued pseudodifferential operator of order  $-1$ , and on a neighborhood of  $\text{supp } \eta$ ,

$$(20) \quad q_{-1}(x, \xi) = \|\xi\|^{-1} I_{\mathbb{C}^G}.$$

We use the Kohn–Nirenberg convention

$$\text{Op}(a)u(x) = (2\pi)^{-N} \int_{\mathbb{R}^{2N}} e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy d\xi.$$

If  $a$  and  $b$  are classical symbols, the composition formula [?, Theorem 18.1.8, p. 71] gives

$$(21) \quad a \# b \sim \sum_{\beta \in \mathbb{N}_0^N} \frac{1}{\beta!} \partial_\xi^\beta a D_x^\beta b, \quad D_x := \frac{1}{i} \partial_x.$$

Write, on a neighborhood of  $\text{supp } \eta$ ,

$$\sigma(Q_\psi)(x, \xi) \sim q_{-1}(x, \xi) + q_{-2}(x, \xi) + \cdots.$$

By (4),

$$\sigma(\tilde{P})(x, \xi) \sim p_2(x, \xi) + p_1(x, \xi) + p_0(x, \xi),$$

where

$$p_2(x, \xi) = \|\xi\|^2 I_{\mathbb{C}^G}, \quad p_1(x, \xi) = 0.$$

Since  $\tilde{Q} = \tilde{P}^{-1/2}$ ,  $\tilde{Q}\tilde{P}\tilde{Q} = I$ . Applying (21) on the region where  $\psi = 1$  and comparing the homogeneous terms of degree  $-1$ , the terms containing  $x$ -derivatives of  $p_2$  and  $q_{-1}$  vanish, and hence

$$q_{-2}p_2q_{-1} + q_{-1}p_2q_{-2} = 0.$$

Together with (20), this gives

$$(22) \quad q_{-2}(x, \xi) = 0$$

on a neighborhood of  $\text{supp } \eta$ .

Applying (21) to the commutator  $[Q_\psi, M_F]$ , and using (20)–(22), we obtain, on the same neighborhood,

$$\begin{aligned} \sigma_{-2}([Q_\psi, M_F])(x, \xi) &= \frac{1}{i} \sum_{\ell=1}^N \partial_{\xi_\ell} (\|\xi\|^{-1}) \partial_{x_\ell} F(x) \\ (23) \quad &= i \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Since  $\psi = 1$  on a neighborhood of  $\operatorname{supp} \eta$  and  $\eta = 1$  on a neighborhood of  $\operatorname{supp} \chi$ ,

$$\begin{aligned} M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi &= M_\chi[\tilde{T}_j, M_F] Q_\psi M_\chi, \\ M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi &= M_\chi \tilde{T}_j M_\eta [Q_\psi, M_F] M_\chi. \end{aligned}$$

Both operators on the right are classical matrix valued pseudodifferential operators of order  $-1$ .

By (19), (20), and the order-zero part of the composition formula,

$$\begin{aligned} \sigma_{-1}(M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g(x) \rangle) \\ (24) \quad &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

Under the above Kohn–Nirenberg convention, (3) gives

$$\sigma_1(\tilde{T}_j)(x, \xi) = i \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle).$$

Therefore, by (23),

$$\begin{aligned} \sigma_{-1}(M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi)(x, \xi) \\ (25) \quad &= -\chi(x)^2 \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle \langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Combining (24) and (25), and setting

$$\theta := \frac{\xi}{\|\xi\|},$$

we obtain

$$\begin{aligned} \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left( \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha \\ &= \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}(x, \theta). \end{aligned}$$

This proves (17).  $\square$

**Lemma 8.5.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Then*

$$\left( \int_\Gamma \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* By the definition of  $\mathbf{c}_{f,j}$ , the unitary invariance of the Schatten norm, and  $\|d_{g,j}\| = 1$ ,

$$\begin{aligned} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N} &\lesssim_{R,k} \left( \sum_{g \in G} |\nabla f_g(x)|^N \right)^{1/N} \\ &\quad + \sum_{\alpha \in R_+} \left( \sum_{g \in G} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N \right)^{1/N}. \end{aligned}$$

The chamber decomposition used in Lemma 8.3 and the orthogonality of  $g \in G$  imply

$$\sum_{g \in G} \int_{\Gamma} |\nabla f_g(x)|^N dx = \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N.$$

Moreover, since  $\|\alpha\|^2 = 2$ ,

$$|x - \sigma_\alpha x| = \sqrt{2} |\langle \alpha, x \rangle|,$$

and hence

$$\left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right| = \sqrt{2} \mathcal{G}_{g, g\sigma_\alpha} f(x)$$

for almost every  $x \in \mathbb{R}^N$ . By Lemma ??,

$$\int_{\Gamma} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N dx \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N$$

uniformly in  $g \in G$  and  $\alpha \in R_+$ . Since  $G$  and  $R_+$  are finite,

$$(26) \quad \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \lesssim_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

For the converse estimate, let  $P_g$  denote the orthogonal projection of  $\mathbb{C}^G$  onto the  $g$ -th coordinate and define

$$\mathcal{E}(A) := \sum_{g \in G} P_g A P_g, \quad A \in \mathcal{L}(\mathbb{C}^G).$$

Thus,  $\mathcal{E}(A)$  is the diagonal part of  $A$ . The pinching inequality for unitarily invariant norms [?, p. 97, (IV.51)–(IV.52)] gives

$$(27) \quad \|\mathcal{E}(A)\|_{S^N} \leq \|A\|_{S^N}.$$

For every  $\alpha \in R_+$ ,  $S_\alpha$  has zero diagonal, since  $g\sigma_\alpha \neq g$  for all  $g \in G$ . Hence, by the definition of  $\mathbf{c}_{f,j}$ ,

$$\mathcal{E}(\mathbf{c}_{f,j}(x, \theta)) = \text{diag}_{g \in G} \left( \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right).$$

Therefore, by (27),

$$(28) \quad \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N \geq \sum_{g \in G} \left| \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right|^N.$$

For each  $g \in G$ , set

$$y := gx, \quad \eta := g\theta.$$

Since

$$\nabla f_g(x) = g^{-1} \nabla f(y), \quad d_{g,j} = g^{-1} e_j,$$

and  $g$  is orthogonal,

$$\begin{aligned} \langle d_{g,j}, \nabla f_g(x) \rangle &= \partial_j f(y), \\ \langle d_{g,j}, \theta \rangle &= \eta_j, \\ \langle \theta, \nabla f_g(x) \rangle &= \langle \eta, \nabla f(y) \rangle. \end{aligned}$$

Using the chamber decomposition and the orthogonal invariance of surface measure, (28) therefore gives

$$\begin{aligned} (29) \quad & \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N d\theta dx \\ & \geq \int_{\mathbb{R}^N} \int_{\mathbb{S}^{N-1}} |\partial_j f(y) - \eta_j \langle \eta, \nabla f(y) \rangle|^N d\eta dy. \end{aligned}$$

For  $v \in \mathbb{C}^N$ , set

$$\Phi_j(v) := \left( \int_{\mathbb{S}^{N-1}} |v_j - \eta_j \langle \eta, v \rangle|^N d\eta \right)^{1/N}.$$

Then

$$(30) \quad \Phi_j(v) \simeq_N \|v\|.$$

Indeed,  $\Phi_j$  is a seminorm on  $\mathbb{C}^N$ . If  $\Phi_j(v) = 0$ , continuity gives

$$v_j - \eta_j \langle \eta, v \rangle = 0, \quad \eta \in \mathbb{S}^{N-1}.$$

Since  $N > 1$ , there exists  $\eta \in \mathbb{S}^{N-1}$  with  $\eta_j = 0$ , and therefore  $v_j = 0$ . It follows that

$$\langle \eta, v \rangle = 0$$

whenever  $\eta_j \neq 0$ . Since such  $\eta$  form a dense subset of  $\mathbb{S}^{N-1}$ , continuity yields  $\langle \eta, v \rangle = 0$  for every  $\eta \in \mathbb{S}^{N-1}$ , and hence  $v = 0$ . Thus  $\Phi_j$  is a norm, and (30) follows from the equivalence of norms on the finite dimensional space  $\mathbb{C}^N$ .

Applying (30) with  $v = \nabla f(y)$  in (29), we obtain

$$\|\nabla f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_N \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Combining this estimate with (26) proves the assertion.  $\square$

**Lemma 8.6.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in C_c^\infty(\mathbb{R}^N)$ . Then*

$$(31) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

*Proof.* Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

By Lemma ?? (??), there exist  $\chi_m \in C_c^\infty(\Gamma_+)$ ,  $0 \leq \chi_m \leq 1$ , such that  $\chi_m(x) \rightarrow 1$  for every  $x \in \Gamma_+$  and

$$(32) \quad \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \rightarrow 0.$$

Set

$$A_{f,m} := M_{\chi_m} A_f M_{\chi_m}, \quad K_m := \text{supp } \chi_m.$$

By Lemma 8.4 (17),  $A_{f,m}$  is a compactly supported classical matrix valued pseudodifferential operator of order  $-1$  on  $\Gamma_+$ , with

$$(33) \quad \sigma_{-1}(A_{f,m})(x, \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left( x, \frac{\xi}{\|\xi\|} \right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Choose a bounded open set  $U_m$  with  $K_m \Subset U_m \Subset \Gamma_+$ , and choose  $L_m > 0$  such that

$$L_m > \sup_{x,y \in U_m} \|x - y\|_\infty.$$

For

$$\mathbb{T}_m^N := \mathbb{R}^N / L_m \mathbb{Z}^N,$$

the quotient map restricts to a diffeomorphism

$$\kappa_m : U_m \rightarrow \Omega_m := \kappa_m(U_m),$$

which is a local isometry. Since the Schwartz kernel of  $A_{f,m}$  is supported in  $K_m \times K_m$ , with respect to

$$L^2(\Gamma; \mathbb{C}^G) = L^2(U_m; \mathbb{C}^G) \oplus L^2(\Gamma \setminus U_m; \mathbb{C}^G)$$

we have

$$A_{f,m} = B_{f,m} \oplus 0$$

for some  $B_{f,m}$  on  $L^2(U_m; \mathbb{C}^G)$ .

Let

$$W_m : L^2(U_m; \mathbb{C}^G) \rightarrow L^2(\Omega_m; \mathbb{C}^G), \quad W_m u := u \circ \kappa_m^{-1}.$$

Then  $W_m$  is unitary. By the invariance of classical pseudodifferential operators under smooth coordinate changes [?, Theorem 18.1.17],

$$T_{f,m} := W_m B_{f,m} W_m^{-1} \in \Psi_{\text{cl}}^{-1}(\Omega_m; \mathbb{C}^G).$$

Its Schwartz kernel is supported in  $\kappa_m(K_m) \times \kappa_m(K_m) \Subset \Omega_m \times \Omega_m$ . Choose  $\vartheta_m \in C_c^\infty(\Omega_m)$  equal to 1 on a neighborhood of  $\kappa_m(K_m)$ . Then

$$T_{f,m} = M_{\vartheta_m} T_{f,m} M_{\vartheta_m}.$$

Hence the zero extension of the Schwartz kernel of  $T_{f,m}$  from  $\Omega_m \times \Omega_m$  to  $\mathbb{T}_m^N \times \mathbb{T}_m^N$  is a classical pseudodifferential kernel of order  $-1$ : locally in  $\Omega_m \times \Omega_m$  this follows from  $T_{f,m} \in \Psi_{\text{cl}}^{-1}$ , while near the boundary of  $\Omega_m \times \Omega_m$  the kernel vanishes. Denote the corresponding operator on  $\mathbb{T}_m^N$  by  $\hat{A}_{f,m}$ . Thus,

$$(34) \quad \hat{A}_{f,m} \in \Psi_{\text{cl}}^{-1}(\mathbb{T}_m^N; \mathbb{C}^G).$$

Since  $\kappa_m$  is a local isometry, by (33) and the transformation law for principal symbols,

$$(35) \quad \sigma_{-1}(\hat{A}_{f,m})(\kappa_m(x), \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left( x, \frac{\xi}{\|\xi\|} \right), \quad x \in U_m, \quad \xi \neq 0,$$

and the principal symbol vanishes outside  $\Omega_m$ .

With respect to

$$L^2(\mathbb{T}_m^N; \mathbb{C}^G) = L^2(\Omega_m; \mathbb{C}^G) \oplus L^2(\mathbb{T}_m^N \setminus \Omega_m; \mathbb{C}^G),$$

we have

$$\widehat{A}_{f,m} = T_{f,m} \oplus 0.$$

Since  $T_{f,m}$  and  $B_{f,m}$  are unitarily equivalent,

$$(36) \quad \mu(t, \widehat{A}_{f,m}) = \mu(t, A_{f,m}), \quad t > 0.$$

By (34) and the Weyl law [?, Theorem 6.1],

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, \widehat{A}_{f,m}) = \left( \frac{1}{N(2\pi)^N} \int_{S^* \mathbb{T}_m^N} \text{tr}_{\mathbb{C}^G} |\sigma_{-1}(\widehat{A}_{f,m})(z, \theta)|^N dz d\theta \right)^{1/N}.$$

Since the metric on  $\mathbb{T}_m^N$  is flat, combining (35) with (36), we obtain

$$(37) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_{f,m}) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N},$$

where  $\chi_m$  is identified with its zero extension to  $\Gamma$ .

By Lemma 8.5,

$$(x, \theta) \mapsto \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N$$

belongs to  $L^1(\Gamma \times \mathbb{S}^{N-1})$ . Since  $0 \leq \chi_m \leq 1$ ,  $\chi_m \rightarrow 1$  on  $\Gamma_+$ , and  $\Gamma \setminus \Gamma_+$  is contained in finitely many hyperplanes, the dominated convergence theorem implies

$$(38) \quad \begin{aligned} & \lim_{m \rightarrow \infty} \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \\ &= \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}. \end{aligned}$$

By (32),  $A_{f,m} \rightarrow A_f$  in  $S^{N,\infty}$ . Combining (37), (38), and Lemma 8.7, we obtain

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_f) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Since  $U$  and  $V$  are unitary,

$$\mu(t, A_f) = \mu(t, [R_{\text{Dunkl},j}, M_f]), \quad t > 0.$$

This completes the proof of Lemma 8.6.  $\square$

**Lemma 8.7.** *Let  $0 < p < \infty$ , let  $\mathcal{H}$  be a Hilbert space, and let  $A, A_m \in S^{p,\infty}(\mathcal{H})$ ,  $m \geq 1$ , satisfy*

$$\|A_m - A\|_{S^{p,\infty}} \longrightarrow 0.$$

*Suppose that, for every  $m \geq 1$ , the limit*

$$c_m := \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A_m)$$

*exists. Then both limits*

$$\lim_{m \rightarrow \infty} c_m \quad \text{and} \quad \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A)$$

exist and

$$\lim_{t \rightarrow \infty} t^{1/p} \mu(t, A) = \lim_{m \rightarrow \infty} c_m.$$

*Proof.* The assertion is the Birman–Solomyak approximation lemma; see [?, Section 11.6].  $\square$

**Lemma 8.8.** *Assume that  $N > 1$ ,  $j \in \{1, \dots, N\}$ , and  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ . Then there exists  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that*

$$\begin{aligned} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} &\longrightarrow 0, \\ \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &\longrightarrow 0, \\ \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &\longrightarrow 0. \end{aligned}$$

*Proof.* By the density of  $C_c^\infty(\mathbb{R}^N)$  in  $\dot{W}^{1,N}(\mathbb{R}^N)$ , choose  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that

$$\|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0.$$

By Theorem ?? and Lemma ??,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &= \|[R_{\text{Dunkl},j}, M_{f_m-f}]\|_{S^{N,\infty}} \\ &\lesssim \|f_m - f\|_{OSC^{N,\infty}(\mathbb{R}^N)} \lesssim \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

By the linearity of  $f \mapsto \mathfrak{c}_{f,j}$  and Lemma 8.5,

$$\begin{aligned} \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &= \|\mathfrak{c}_{f_m-f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \\ &\lesssim_{N,R,k} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

This completes the proof of Lemma 8.8.  $\square$

**Theorem 8.9.** *Assume that  $N > 1$  and  $j \in \{1, \dots, N\}$ . Then, for every  $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ ,*

(39)

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left( \frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Moreover,

$$(40) \quad \left( \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

*Proof.* By Lemma 8.8, choose  $f_m \in C_c^\infty(\mathbb{R}^N)$  such that

$$(41) \quad \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \longrightarrow 0,$$

$$(42) \quad \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow 0.$$

By Lemma 8.6, for every  $m$ ,

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_{f_m}]) = \frac{1}{(N(2\pi)^N)^{1/N}} \|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

By (42),

$$\|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow \|\mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

Combining this convergence with (41), we apply Lemma 8.7 with  $p = N$  and obtain (39). Finally, (40) follows from Lemma 8.5. This completes the proof of Theorem 8.9.  $\square$

## APPENDIX A. ZHIJIE LEMMA

**Lemma A.1.** *Let  $1 < s < \infty$ , and let  $\mathbb{D}^0$  be the standard dyadic grid in  $\mathbb{R}^N$ . Suppose that  $F \in L^s(\mathbb{R}^N)$ . Then there exists a constant  $C > 0$  such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

*Proof.* Let  $\mathcal{E} \subset \mathbb{D}^0$  be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed  $x$  are totally ordered by inclusion, for each  $x \in \bigcup_{Q \in \mathcal{E}} Q$  we may choose a cube  $Q_x \in \mathcal{E}$  of minimal volume such that  $x \in Q_x$ . Every  $Q \in \mathcal{E}$  containing  $x$  is then a dyadic ancestor of  $Q_x$ . Since the volume increases by the factor  $2^N$  at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix  $t > 0$  and let  $\mathcal{E}$  be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since  $\mathcal{E}$  is arbitrary,

$$\#\left\{ Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t \right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak  $\ell^s$  estimate follows. This completes the proof of Lemma A.1.  $\square$

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